Natural Language Processing

Instructor

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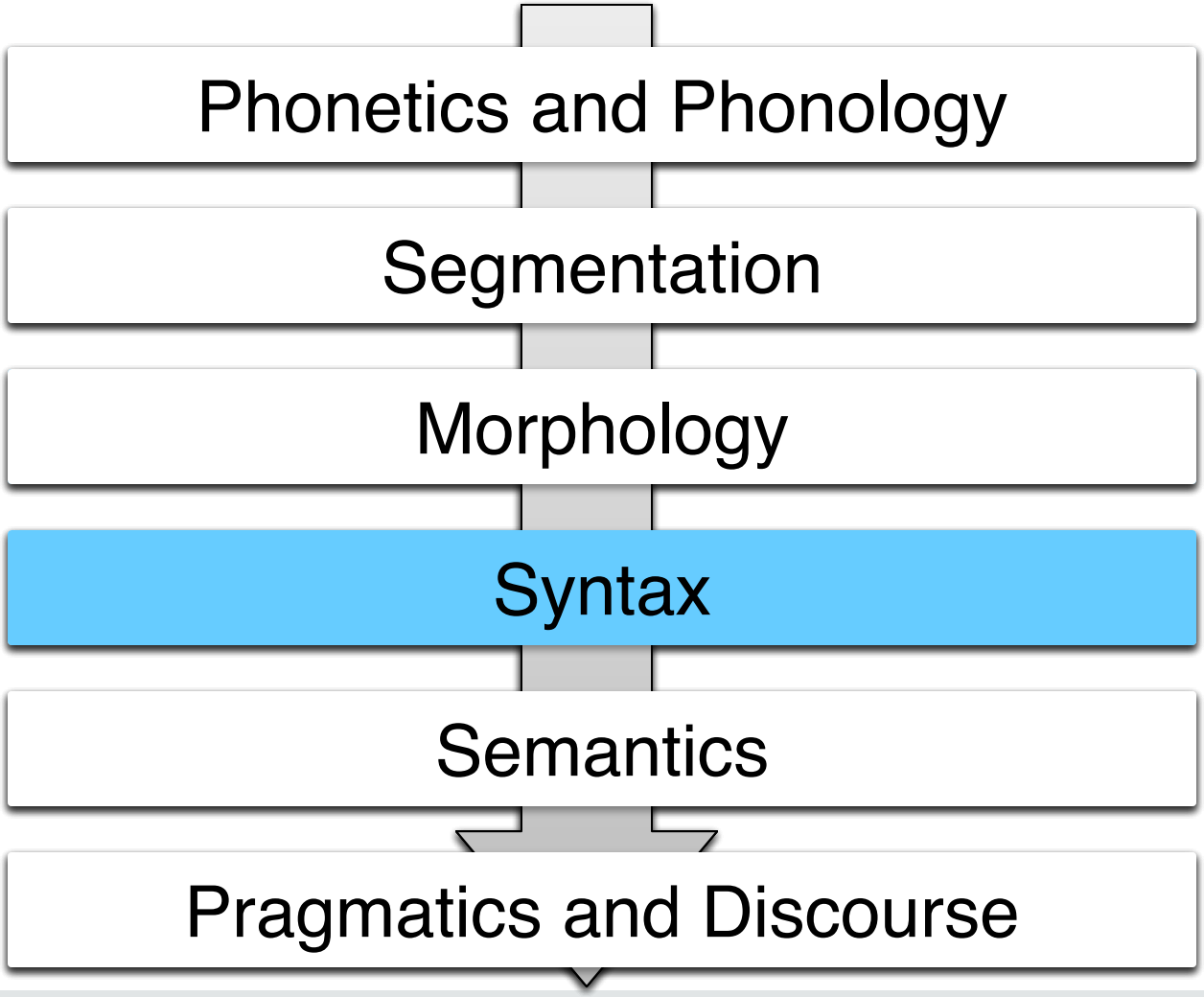
Analysis Levels in Language Understanding



Parts of Speech Tagging: Ambiguities

Problem: Multiple possibilities for assigning parts of speech

- *The* <u>*back*</u> *door* = ADJ
- *On my* <u>*back*</u> = N
- *Win the voters* <u>*back*</u> = ADV
- Promised to <u>*back*</u> the bill = V

Phrases

- **Phrases are contiguous sequences of related words**

  Noun phrases: the president, the Indian president

- They can be identified using standard tests, such as substitution, movement, coordination and question:

  The dog ate a cookie

  The dog ate it (substitution)

  A cookie was eaten by the dog (movement)

  The dog ate a cookie and a sausage (coordination)

  What did the dog eat? A cookie (question)

Heads and Modifiers

- The **head** is the word which determines the syntactic type of the phrase
  - For a noun phrase, the head is the noun

- **Modifiers** qualify another word or phrase
  - In English, adjectives modify nouns and pronouns while adverbs modify verbs, adjectives and other adverbs.

Phrase Types

- A group of words functioning as a single unit in the syntax of a sentence
- The central word defining the type of phrase is called **head of the phrase**

- Phrases are classified by the type of head
  - **Noun phrase (NP)** with a noun as head
    - e.g. the black cat
  - **Verb phrase (VP)** with a verb as head
    - e.g. jump up and down
  - **Adjectival phrase (AP)** with an adjective as head
    - e.g. really slow, terribly sad
  - **Adverbial phrase (AdvP)** with an adverb as head
    - e.g. I exercise very regularly, I eat quite healthily.
  - ……

What is a Grammar?

- A **grammar** is a description of the valid structures in a language
- A grammar is defined by means of a set of **rules**:
  - A → bC : “A structure A is composed of a word b followed by a structure C”
  - C → de : “A structure C is composed of a word d followed by a word e”
- Symbols in the grammar are either **terminal symbols** or **non-terminal symbols**
- Terminal symbols cannot be rewritten any further, they correspond to the words in the language
- Non-terminals express clusters or generalizations of terminals

Context Free Grammar

- Modelling via **Context Free Grammars (CFG)**

- G = (V, $\sum$, R, S)
  - V is set of non-terminals
  - $\sum$ is set of terminals
  - S is the start symbol (one of the nonterminals)
  - R is set of production rules of the form X → ***Y***, where X is a nonterminal and ***Y*** is a sequence of terminals and nonterminals
- A grammar G generates a language L

Example Grammar for English

- S → NP VP
- S → VP
- VP → V NP
- VP → V NP PP
- NP → NP PP
- NP → N
- NP → N N
- PP → P NP

N → *cats*
V → *scratch*
N → *claws*
N → *people*
N → *scratch*
P → *with*

- By convention, S is the **start symbol**

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- S → NP V NP
- S → N V NP
- S → N V NP PP
- S → N V NP P NP
- S → N V N P NP
- S → N V N P N
- S → cats V N P N
- S → cats scratch N P N
- S → cats scratch people P N
- S → cats scratch people with N
- S → cats scratch people with claws

What is Parsing?

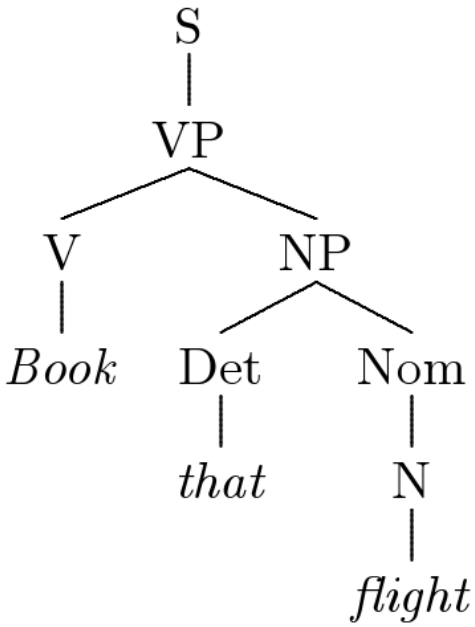
- An automatic analysis of a sentence with respect to its syntactic structure.
- The process of determining the grammatical structure with respect to a given grammar.
- Given a grammar, this means deriving a phrase structure tree assigned to the sentence by the grammar

Representations of Parsing

- Bracketed notation:
  $[_S\ [_{NP}\ [_{Det}$ the$]\ [_N$ dog$]\ ]\ [_{VP}\ [_V$ ate$]\ [_{NP}\ [_{Det}$ a$]\ [_N$ cookie$]\ ]\ ]\ ]$
- Parenthesized notation:

```
(S
    (NP
        (Det the)
        (N dog) )
    (VP
        (V ate)
        (NP
            (Det a)
            (N cookie))))
```

Parse Tree:



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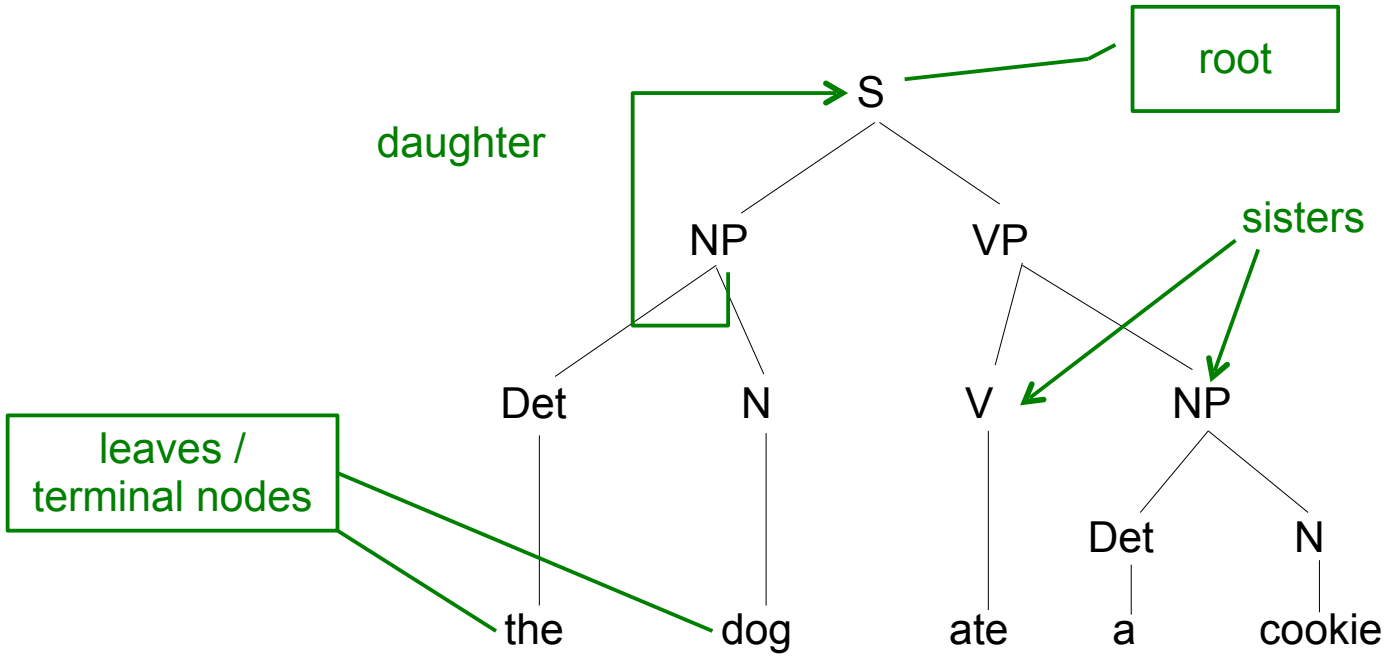
root
S
daughter
NP
VP
sisters
Det
N
V
NP
leaves /
terminal nodes
Det
N
the
dog
ate
a
cookie

Syntactic Ambiguity

- Syntactic ambiguity arises when one sentence has **more than one** syntactic structure

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  - I saw the man with a telescope
  - I shot an elephant in my pyjamas

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  - Black cats and dogs like to play

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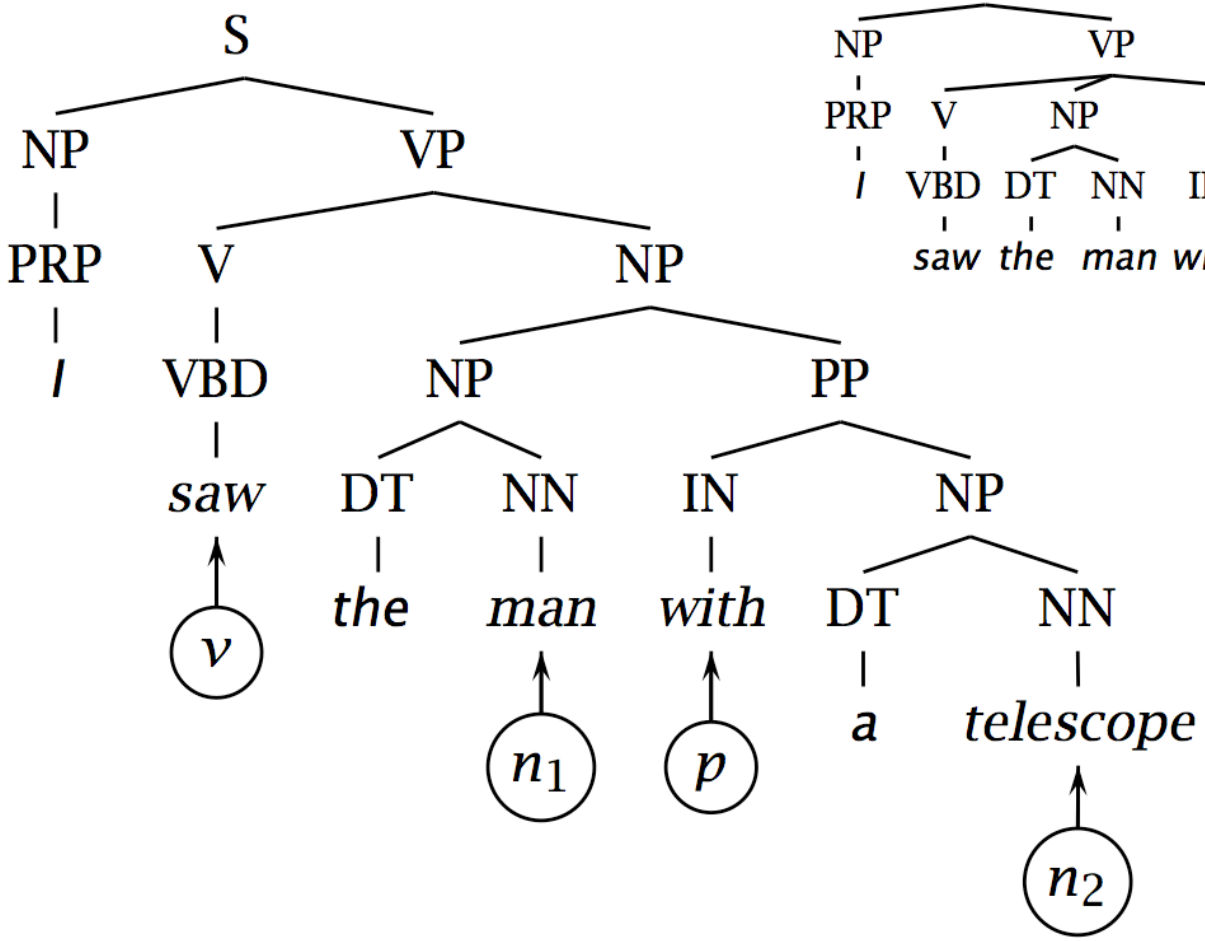
- **Coordination ambiguity:** scope of the conjunction
  - Black cats and dogs like to play

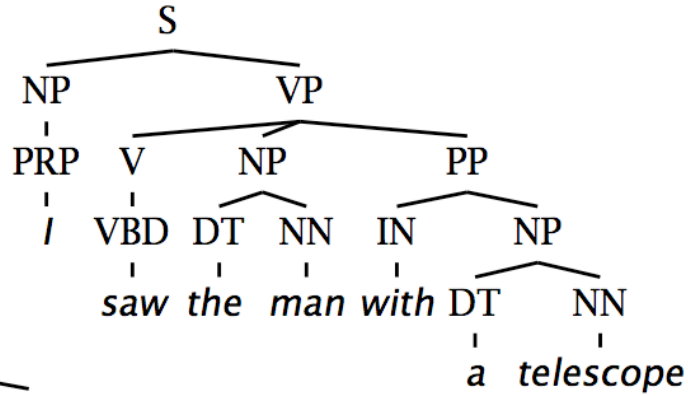
- **Garden path sentences:** sentences which lead you along a path that suddenly turns out not to work
  - The old man the boat.

Attachment ambiguity

"I saw the man with a telescope."

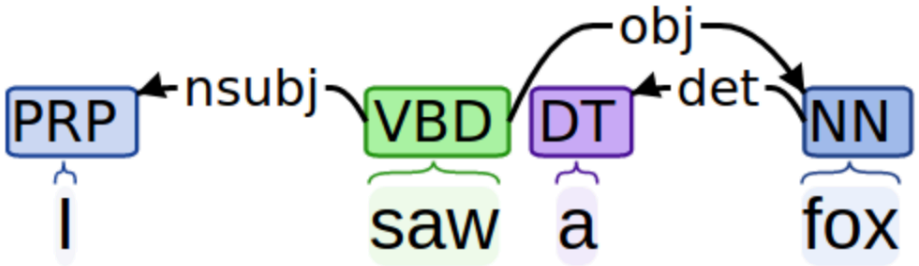
Attachment Ambiguity- Different Parse Trees (Constituency Parsing)





Dependency Parsing

- Dependency structure shows which words depend on (modify or are arguments of) which other words.
- No phrasal/un-lexicalized nodes,
- Syntactic structure of a sentence is described solely in terms of the words



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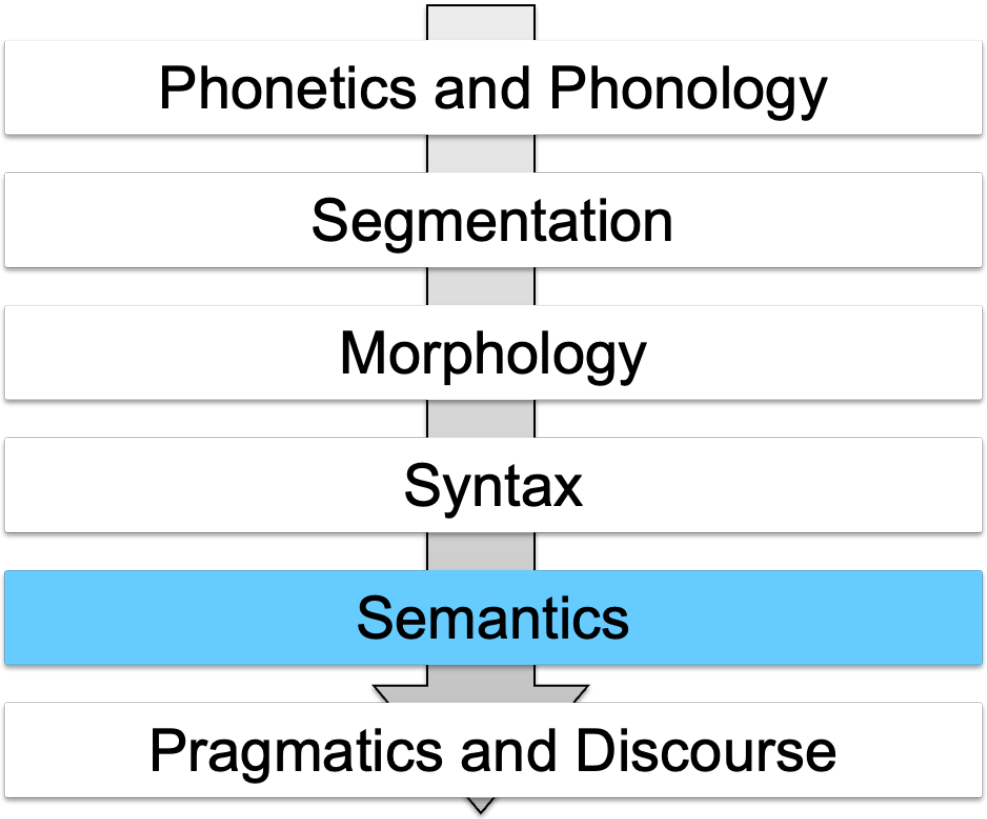


What is Semantics?

- Study of the meaning of **words, phrases, sentences, or documents**

- **Lexical Semantics**
  - Study of the meaning of lexical units, i.e. words

What is the meaning of a word?

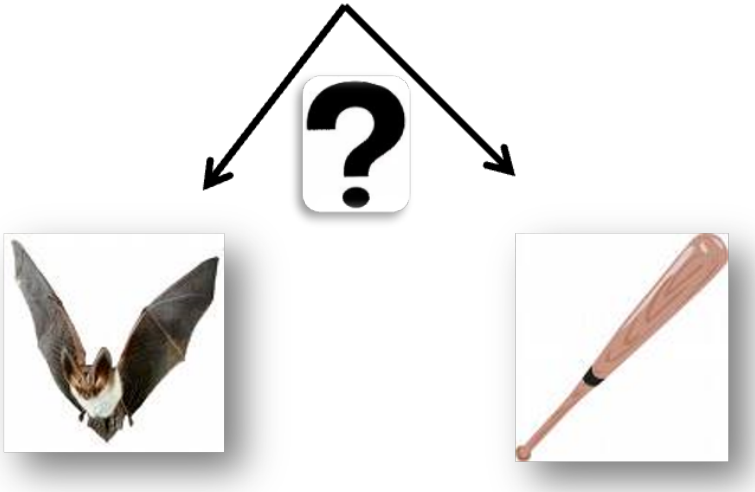
- stone
- unicorn
- greed

Lexical Ambiguity

He hit the ball with the bat.

Lexical Ambiguity

He hit the ball with the bat.



- Different interpretations are caused by lexical ambiguity.

“I made her duck.”

“I made her duck.”

- I cooked duck for her to eat

“I made her duck.”

- I cooked duck for her to eat
- I cooked duck belonging to her

“I made her duck.”

- I cooked duck for her to eat
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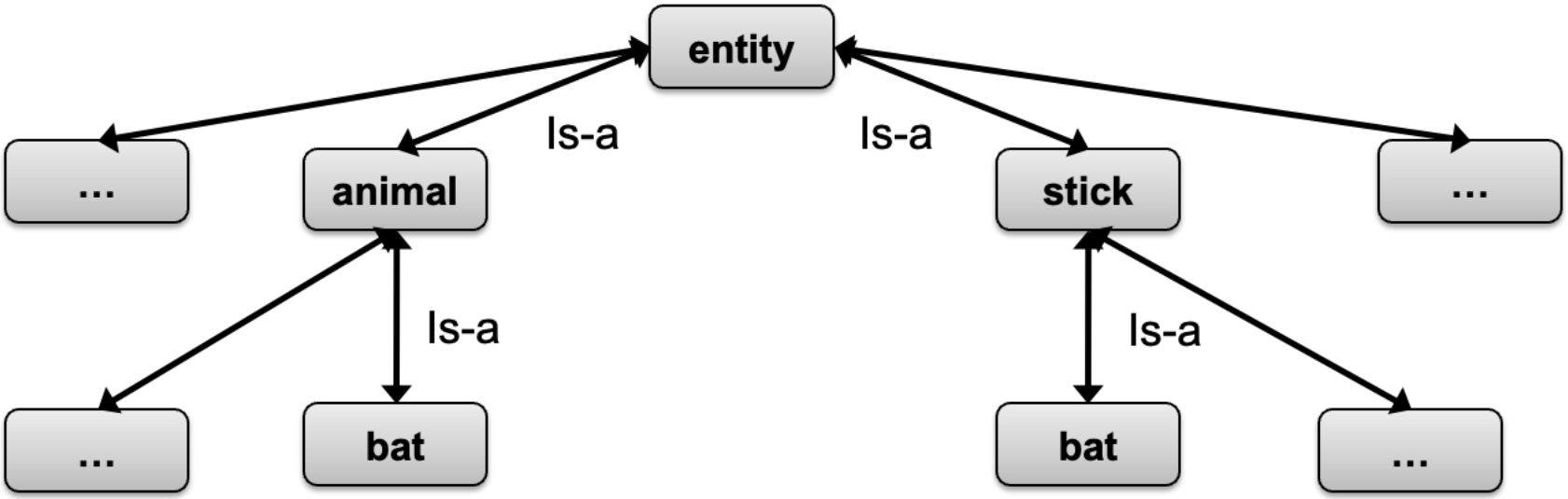
- I created the (plaster?) duck she owns

“I made her duck.”

- I cooked duck for her to eat
- I cooked duck belonging to her
- I created the (plaster?) duck she owns
- I waved my magic wand and turned her into duck

.....

How to Represent Meaning in a Computer?



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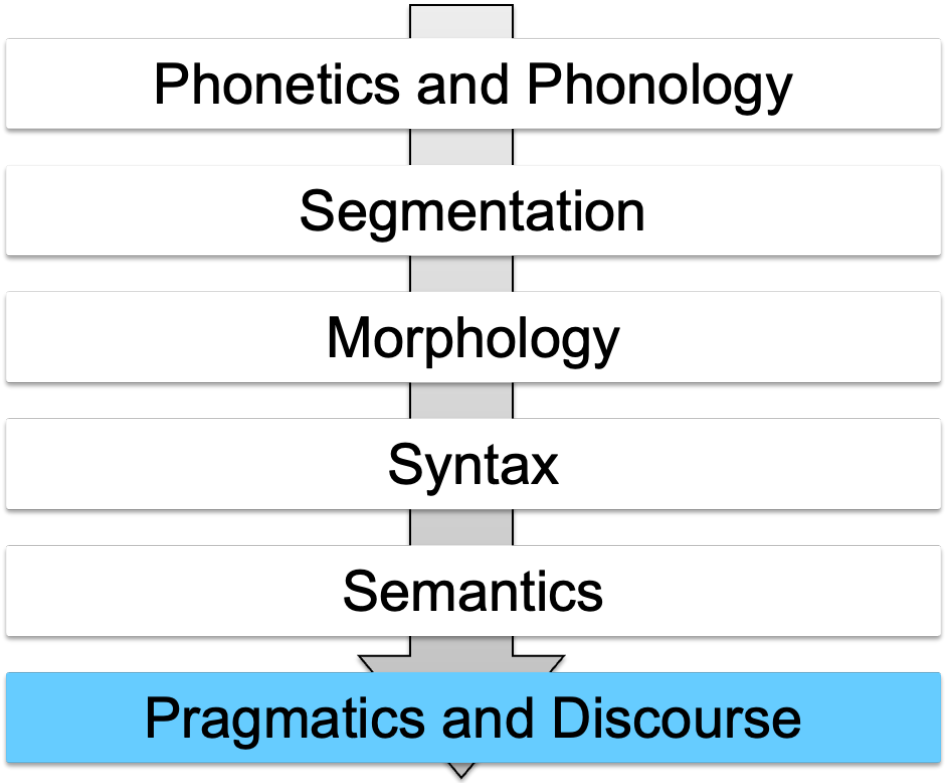

Pragmatics

What is the purpose of an utterance?

Pragmatics

"I never said he stole my money"

Someone else said it, but I didn't.

Pragmatics

"**I** never said he stole my money"

Someone else said it, but I didn't.

"I **never** said he stole my money"

I simply didn't ever say it.

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"I never **said** he stole my money"

I might have implied it in some way, but I never explicitly said it.

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"I never said **he** stole my money"

I said someone took it; I didn't say it was he.

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I just said he probably borrowed it.

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"I never said he stole **my** money"

I said he stole someone else's money.

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I said he stole someone else's money.

"I never said he stole my **money**"

I said he stole something of mine, but not my money.

Discourse Analysis

- Focuses on the structure of naturally spoken language as found in conversation interviews, commentaries, and speech.

- It is an effort to interpret what the writer or speaker intended to convey in a sensitive social context.

Discourse Analysis

- Cohesion
  - ties and connections which exist within texts that link different parts of sentences or larger units of discourse.

"My father once bought a BMW. He bought it by saving every penny he could. That car would be worth a fortune nowadays".

Discourse Analysis

- Coherence
  - unity of ideas, connectedness of Ideas.
  - not something that exists in language but something with exists in people.
  - people try to arrive at an agreement that fits their experience of the world.

“Her: That's the telephone ringing
Him: I’m taking a bath
Her: Ok, I will take the call”

Formal versus Natural Language

Language

- **Language**
  - a set of strings over an alphabet

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{0, 1, 2, 3, 4, 5, 6, 7, 8, 9} the 10-element set of decimal digits.

{a, b, c, …. x, y, z} the 26-element set of lower-case characters of written English.

{aardvark, ….. zebra} the 250,000-element set of words in the Oxford English Dictionary.

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The set of natural numbers N = {0, 1, 2, 3, ….} cannot be an alphabet because it is infinite.

Language

- **Language**
  - a set of strings over an alphabet

Strings: A string of length n over an alphabet $\sum$ is an ordered n-tuple of elements of $\sum$.

$\sum$ * denotes the set of all strings over $\sum$ of finite length.

If $\sum = \{a, b\}$ then ϵ, ba, bab, aab are examples of strings over $\sum$.

If $\sum = \{a\}$ then $\sum^* = \{\epsilon, a, aa, aaa, \ldots.\}$

If $\sum = \{cats, dogs, eat\}$ then

$\sum^* = \{\epsilon, cats, cats\ eat, cats\ eat\ dogs, \ldots..\}$

Given an alphabet Σ, any subset of Σ^* is a language over alphabet Σ.

Formal Language

Noam Chomsky is the originator of the Chomsky hierarchy of formal languages

Why talk about it?

✓ Fundamental results from theoretical computer science have direct implications on the implementation for language technology applications

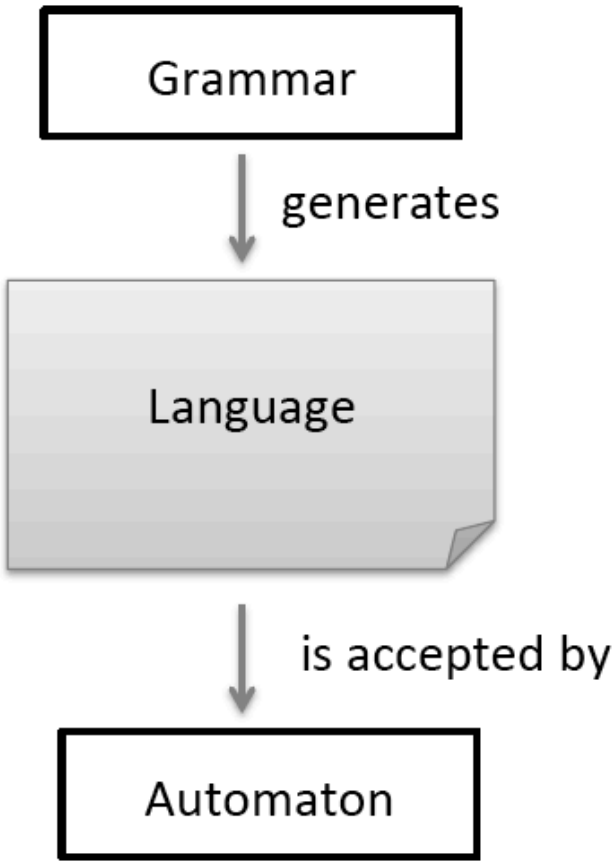
Formal Grammar

- Grammar enumerates the sentence of a language
- A grammar of language L can be regarded as a function whose range is exactly L
- A formal grammar is a quad-tuple G = (N,Σ,S,P) where
  - ✓ N is a finite set of non-terminals
  - ✓ Σ, a finite set of terminals, disjoint from N
  - ✓ S, Element of N: start symbol
  - ✓ P, a finite set of production rules of the form

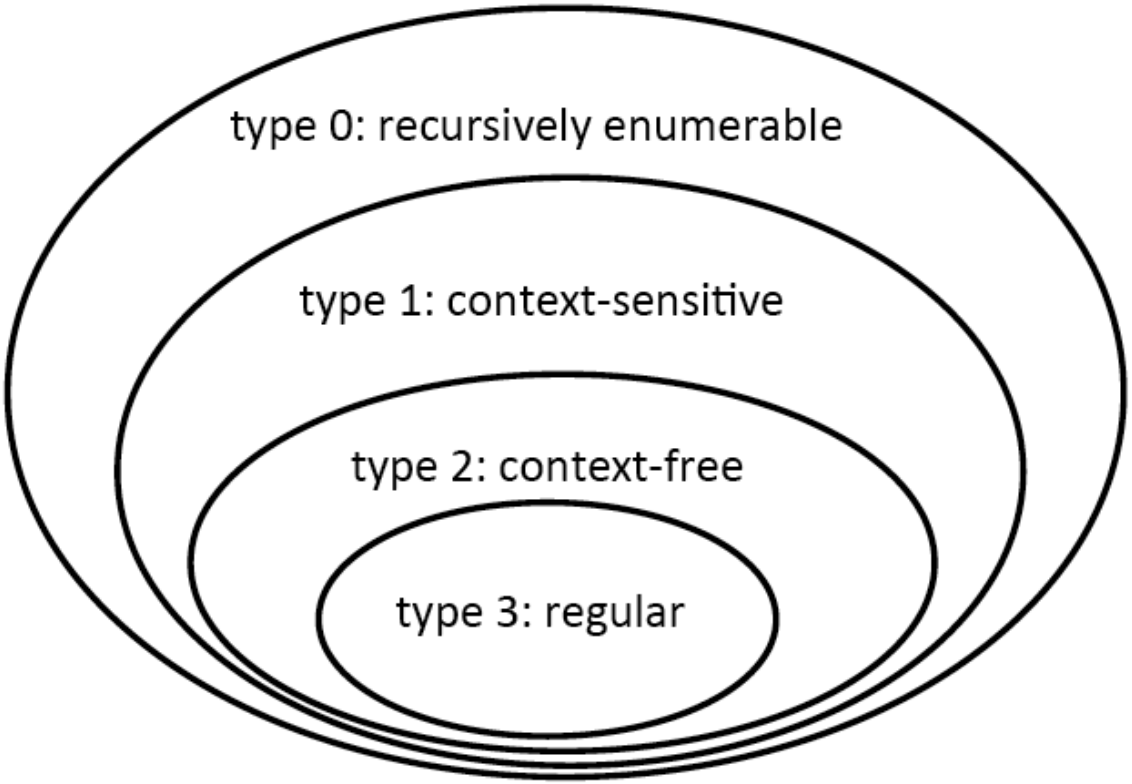
$$\alpha \in (N \cup \Sigma)^* \rightarrow \beta \in (N \cup \Sigma)^* \text{ with } \alpha \neq \varepsilon \text{ and } \alpha \notin \Sigma^*$$

Formal Language

- $G = (N,\Sigma,S,P)$ be a formal grammar.
- Then, $L(G) = \{w \in \Sigma^* \mid S \overset{*}{\Rightarrow} w\}$ is the formal language generated by G.
- An automaton is a device that decides, whether a given sentence belongs to a formal language.



The Chomsky Hierarchy of Formal Languages



Turing Machine (TM)

linearly bounded TM

Pushdown Automaton (PDA)

Finite State Automaton (FSA)

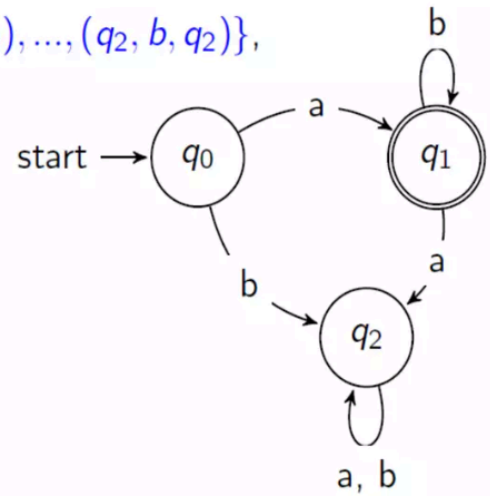
Type 3: Regular Language

A language is regular if it is equal to the set of strings accepted by some deterministic finite-state automaton (DFA).

Regular languages are accepted by DFAs.

For $\mathcal{L}(M) = \{a, ab, abb, \ldots\}$:

$$
\begin{aligned}
M = (\ & \mathcal{Q} = \{q_0, q_1, q_2\}, \\
& \Sigma = \{a, b\}, \\
& \Delta = \{(q_0, a, q_1), (q_0, b, q_2), \ldots, (q_2, b, q_2)\}, \\
& s = q_0, \\
& \mathcal{F} = \{q_1\}\ )
\end{aligned}
$$



Regular Grammar

- Given a DFA M = (Q, Σ, Δ, s, F), the language L(M), of strings accepted by M can be generated by the regular grammar G= (N, Σ, S, P) where:

$N = \{Q\}$ the non-terminals are the states of M

$\sum = \sum$ the terminals, set of transition symbols of M

$S = s$ the starting symbol is the starting state of M

$P = qi \rightarrow aq_j$ when $\delta(qi, a) = qj \in \Delta$

Is the class of Regular grammar expressive enough to model natural language??

- Regular grammars have limitations when modelling natural languages for several reasons
  - Centre Embedding

✓ The students the police arrested complained.

✓ The luggage that the passengers checked arrived.

✓ The luggage that the passengers that the storm delayed checked arrived.

✤ Associated automaton has no memory of what has occurred previously in a string.
✤ To know that n verbs were required to match n nominals already seen, an automaton would need to 'record' that n nominals had been seen;
✤ But a DFA has no mechanism to do this.

Is the class of Regular grammar expressive enough to model natural language??

- Regular grammars have limitations when modelling natural languages for several reasons
  - Redundancy

  ✓ Grammars written using regular grammar rules alone are highly redundant

  ✓ The rules are very simple

  ✓ **We need a great many of them to describe the language.**

  ✓ This makes regular grammars very difficult to build and maintain.

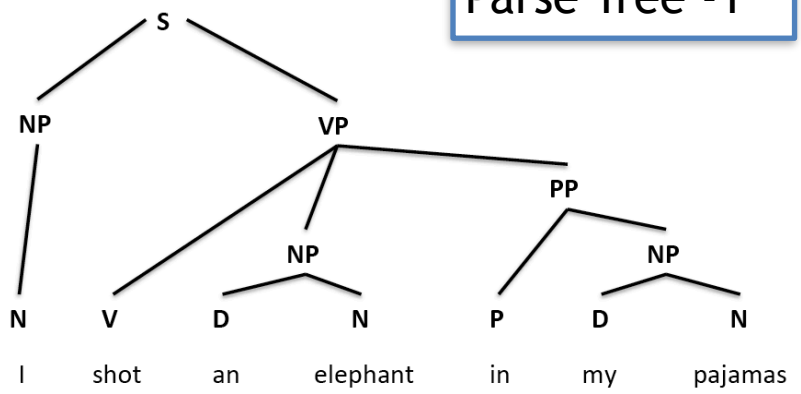
Type 2: Context Free

- A grammar G= (N, Σ, S, P) is context-free, iff all production rules in P obey the form
  - $A \rightarrow \alpha$ with $A \in N$, $\alpha \in (N \cup \Sigma)^*$.
- Context-free grammars are accepted by non-deterministic pushdown automata.

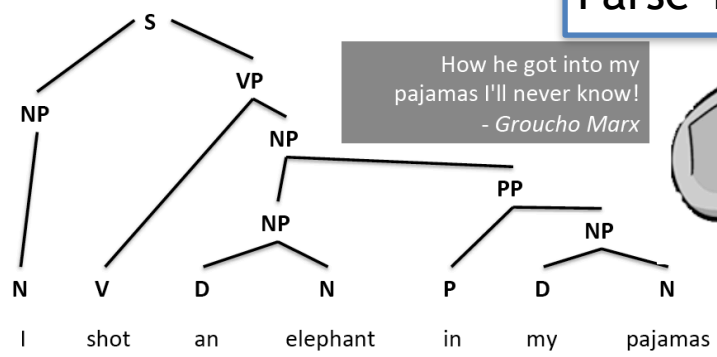
Type 2: Context Free

S	→	NP VP
NP	→	N | D N | NP PP
VP	→	V NP | V NP PP
PP	→	P NP
N	→	I | elephant | pajamas
V	→	shot
P	→	in
D	→	my | an | a | the

Parse Tree -1



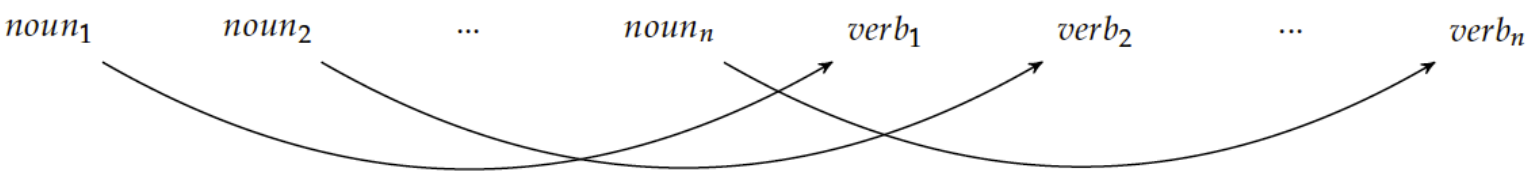
Parse Tree 2



How he got into my pajamas I'll never know!
- Groucho Marx

Is the class of context-free grammar expressive enough to model natural language?

- For every natural language that exists, can we find a context-free grammar to generate it?
- Natural language can contain cross-serial dependencies.

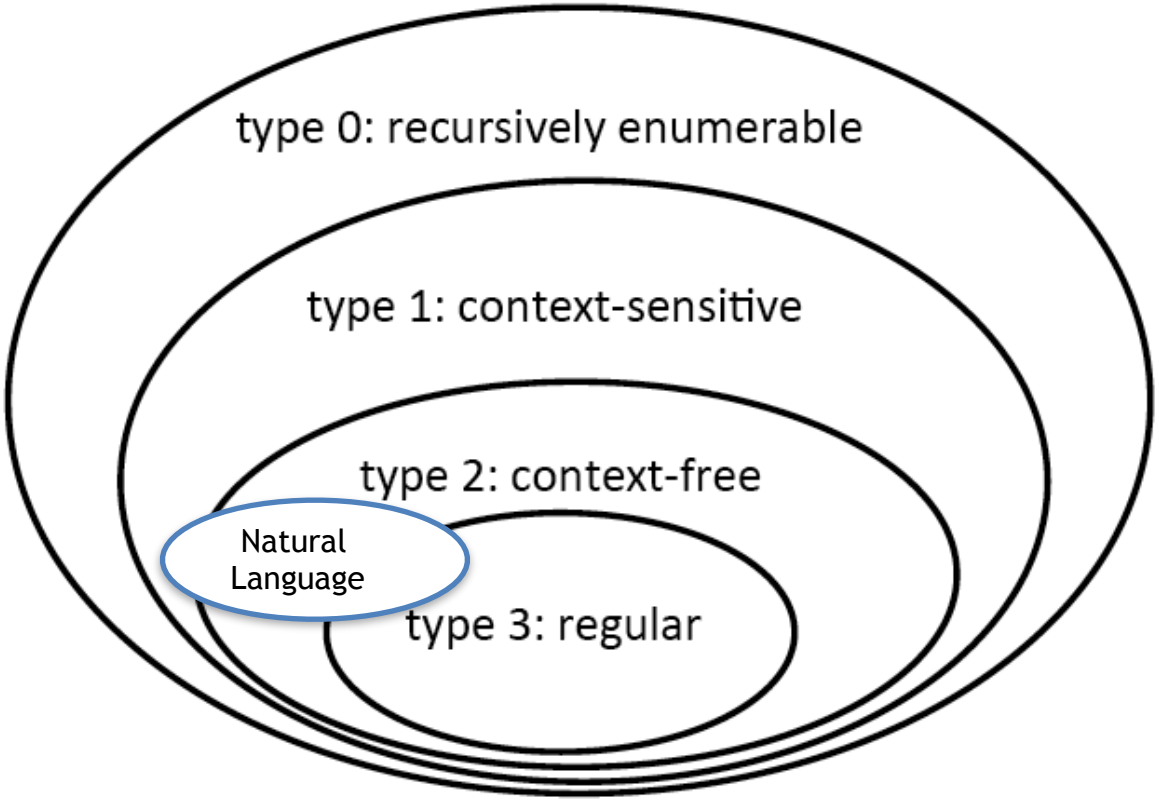


mer d'chind em Hans es huus haend wele laa hälfe aastriiche.

we the children Hans the house have wanted to let help paint.

we have wanted to let the children help Hans paint the house

Where is Natural Language in the Hierarchy ?



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In a nutshell

✓ Natural Languages are an infinite set of sentences constructed out of a finite set of words
✓ Words in a sentence don't have defined upper limits either
✓ Natural languages are believed to be at least context free

✤ Tokenization, morphology ———-> Regular Grammar (mostly)
✤ Syntactic analysis ————-> Context Free Grammar (mostly)
✤ Rest ———> Beyond Formal language modelling